GITHUB ASSISGNNMENT-1
IT23302 DATA STRUCTURES

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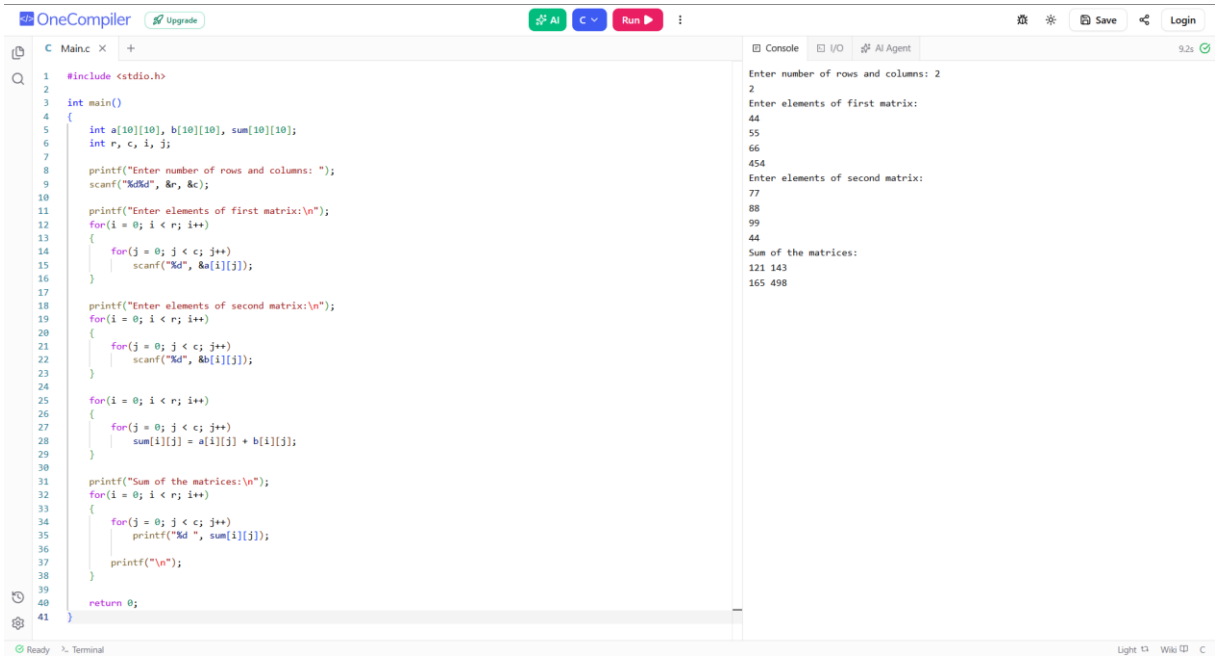
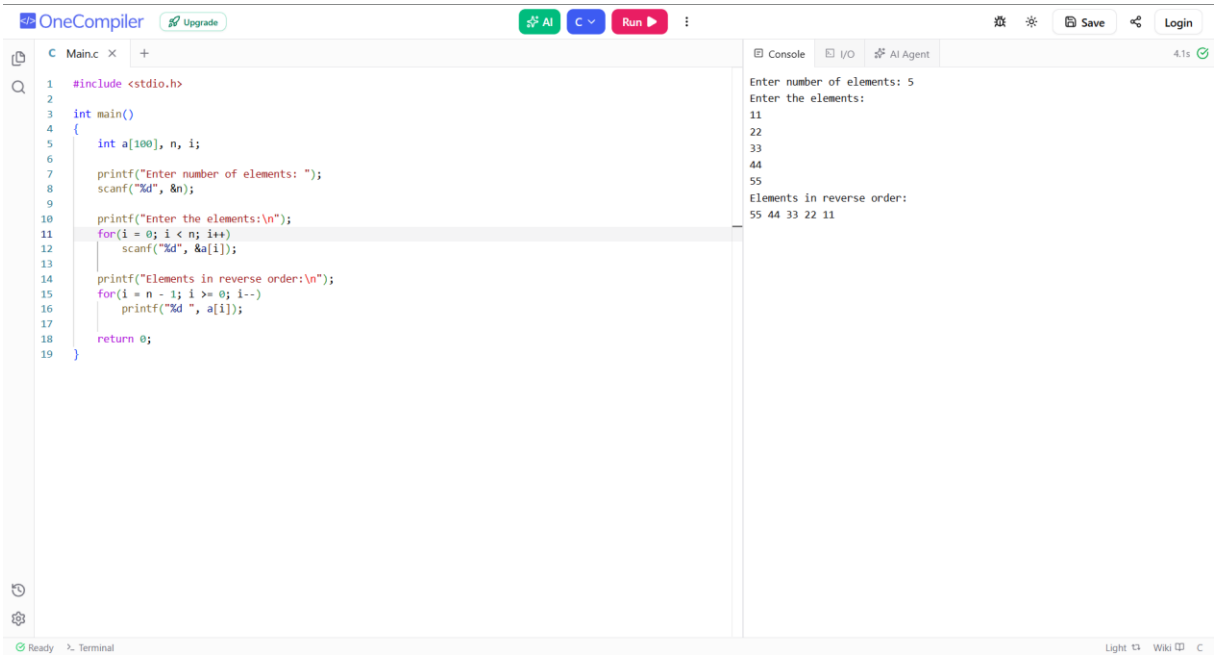
1. Sum of two matrices



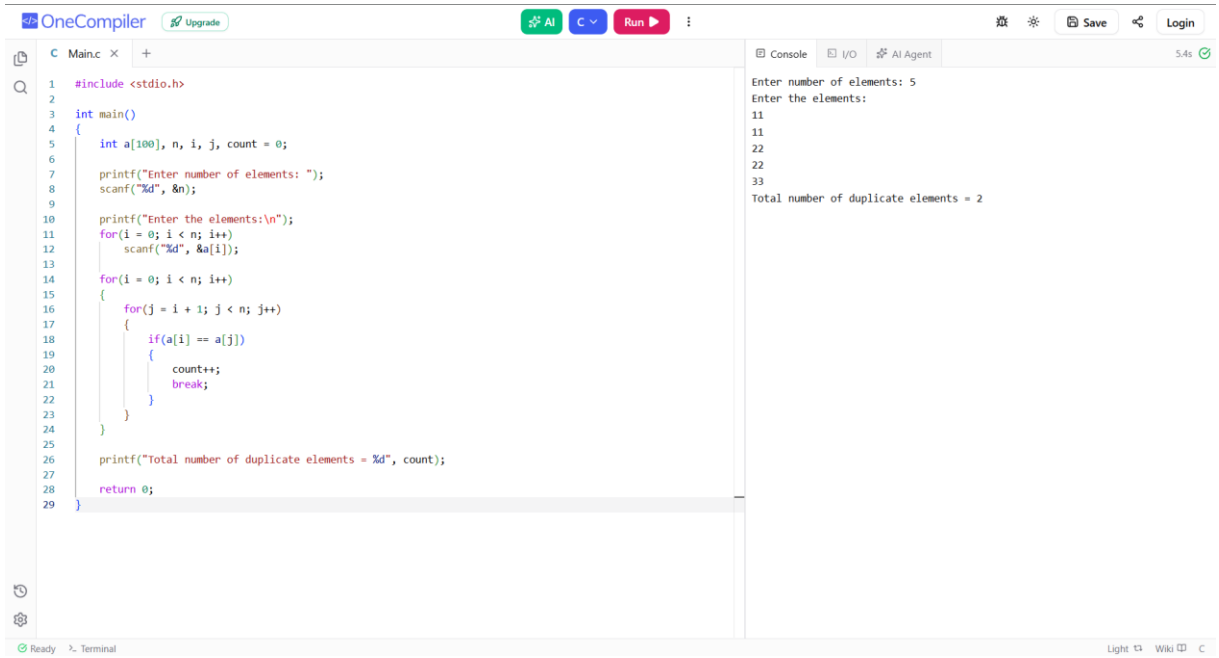
2. Write a C program to read n number of values in an array and display them in reverse order.



```c
#include <stdio.h>

int main()
{
    int a[100], n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Elements in reverse order:\n");
    for(i = n - 1; i >= 0; i--)
        printf("%d ", a[i]);

    return 0;
}
```

```
Enter number of elements: 5
Enter the elements:
11
22
33
44
55
Elements in reverse order:
55 44 33 22 11
```

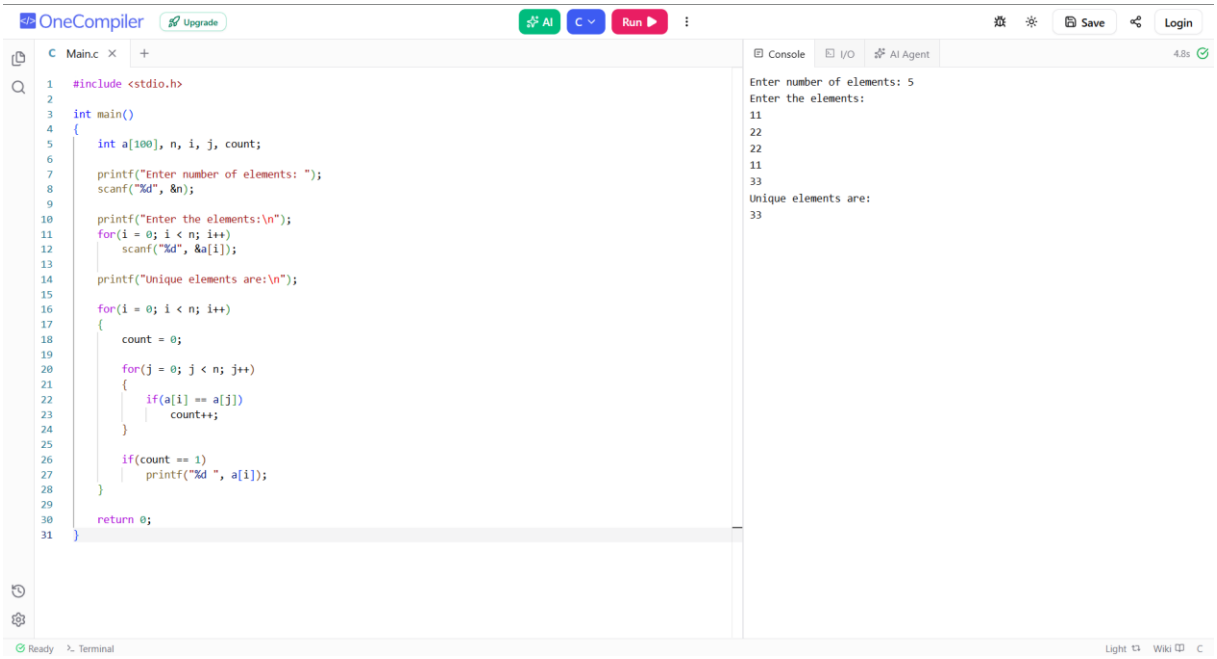
3. Count the total number of duplicate elements in an array.



```c
#include <stdio.h>

int main()
{
    int a[100], n, i, j, count = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total number of duplicate elements = %d", count);

    return 0;
}
```

```
Enter number of elements: 5
Enter the elements:
11
11
22
22
33
Total number of duplicate elements = 2
```

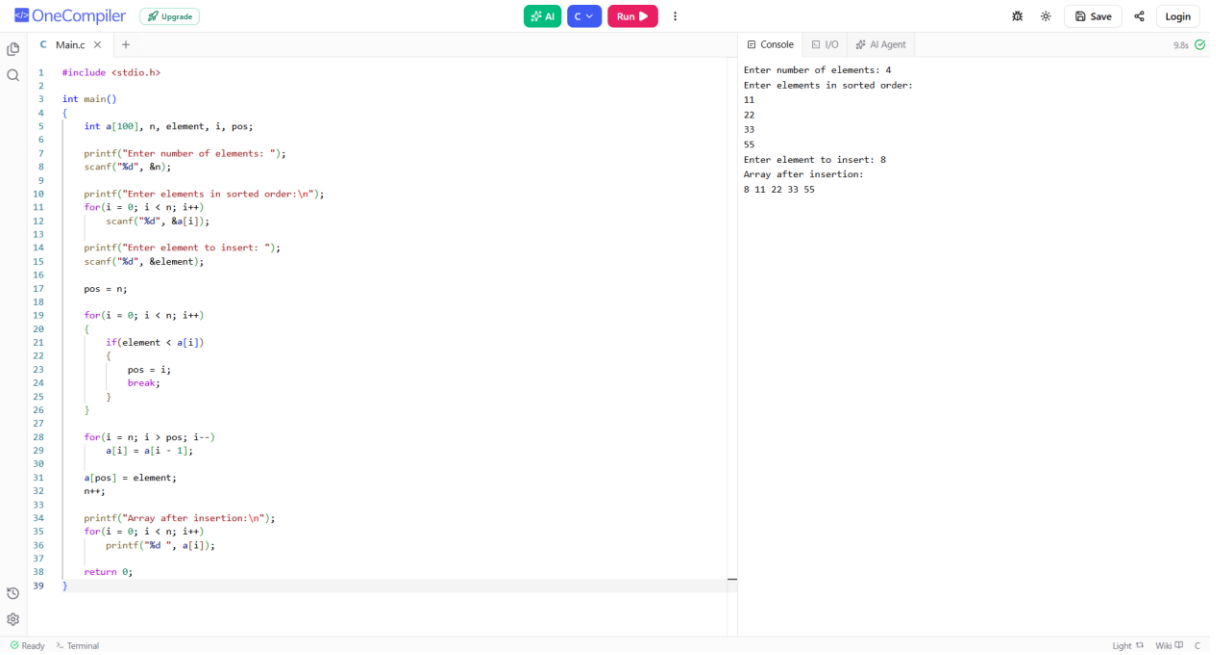
4. Print unique elements in an array:



```c
#include <stdio.h>

int main()
{
    int a[100], n, i, j, count;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Unique elements are:\n");

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(a[i] == a[j])
                count++;
        }

        if(count == 1)
            printf("%d ", a[i]);
    }

    return 0;
}
```

```
Enter number of elements: 5
Enter the elements:
11
22
22
11
33
Unique elements are:
33
```

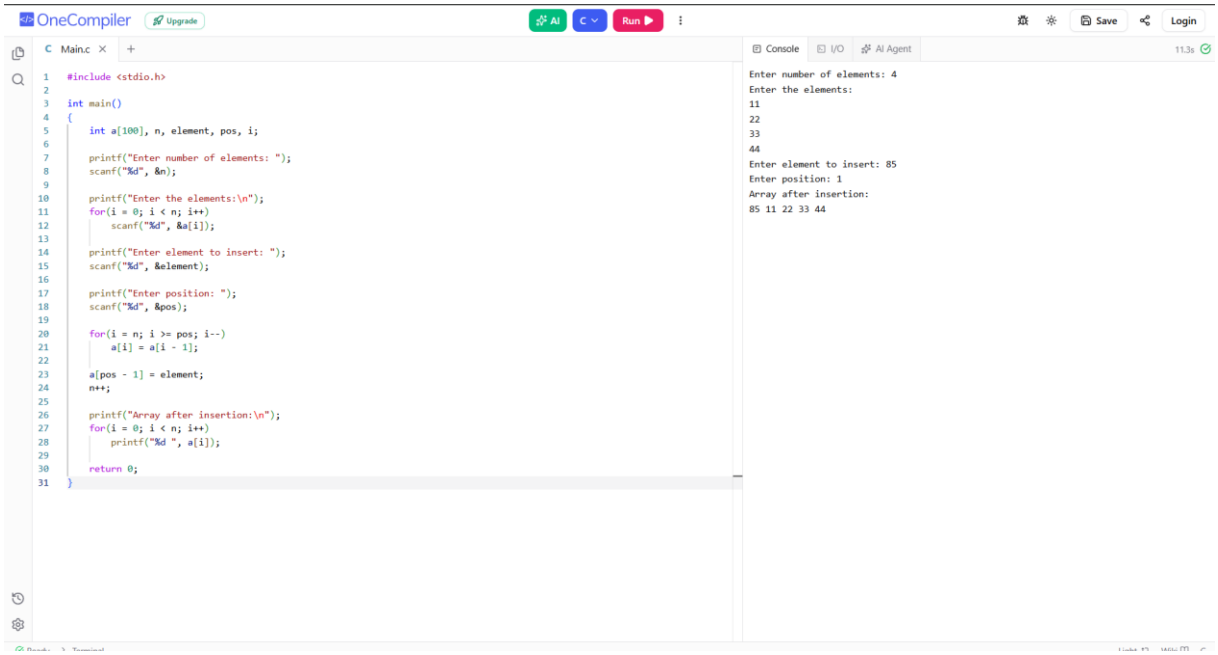
5. Insert in sorted array



```c
#include <stdio.h>

int main()
{
    int a[100], n, element, i, pos;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements in sorted order:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to insert: ");
    scanf("%d", &element);

    pos = n;

    for(i = 0; i < n; i++)
    {
        if(element < a[i])
        {
            pos = i;
            break;
        }
    }

    for(i = n; i > pos; i--)
        a[i] = a[i - 1];

    a[pos] = element;
    n++;

    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

```
Enter number of elements: 4
Enter elements in sorted order:
11
22
33
55
Enter element to insert: 8
Array after insertion:
8 11 22 33 55
```

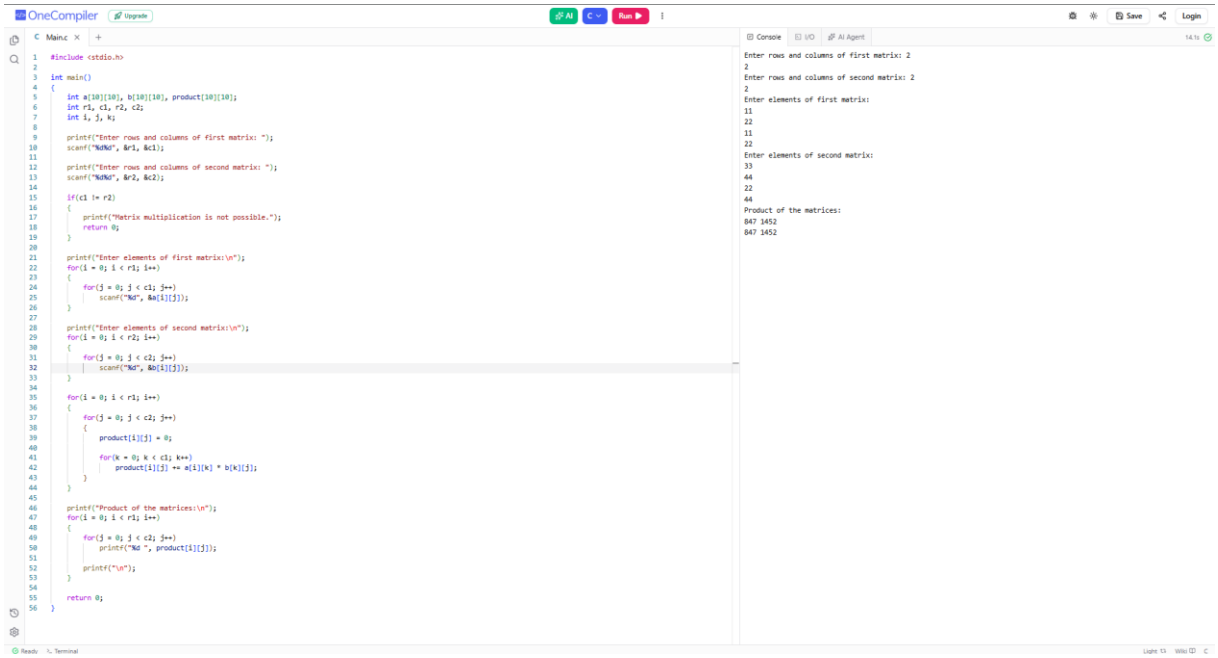
6.Insert in sorted array.



```c
#include <stdio.h>

int main()
{
    int a[100], n, element, pos, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter the elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to insert: ");
    scanf("%d", &element);

    printf("Enter position: ");
    scanf("%d", &pos);

    for(i = n; i >= pos; i--)
        a[i] = a[i - 1];

    a[pos - 1] = element;
    n++;

    printf("Array after insertion:\n");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

```
Enter number of elements: 4
Enter the elements:
11
22
33
44
Enter element to insert: 85
Enter position: 1
Array after insertion:
85 11 22 33 44
```

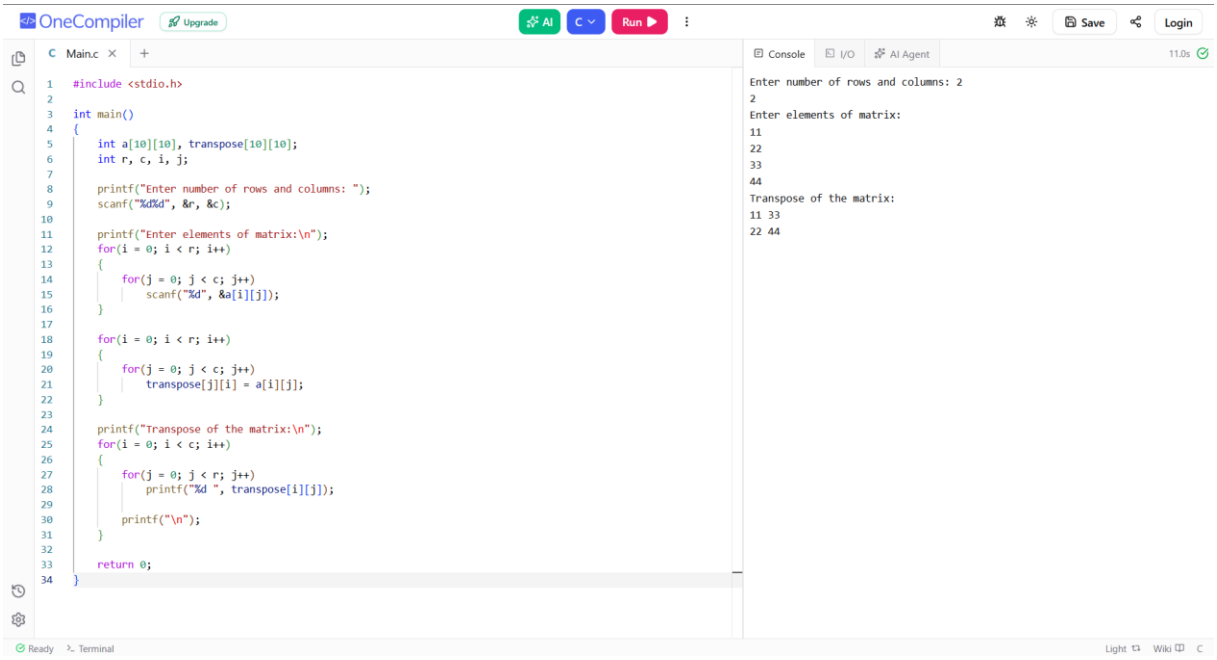
7.Matrix multiplication.



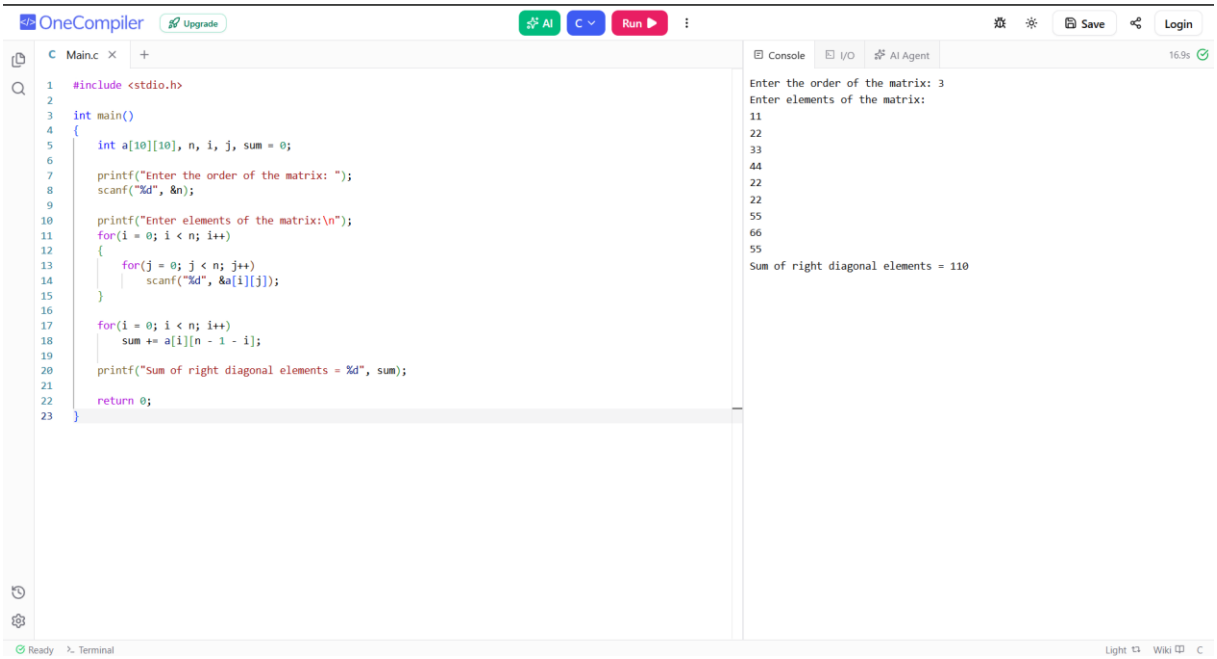
```c
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], product[10][10];
    int r1, c1, r2, c2;
    int i, j, k;

    printf("Enter rows and columns of first matrix: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of second matrix: ");
    scanf("%d%d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Matrix multiplication is not possible.");
        return 0;
    }

    printf("Enter elements of first matrix:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
            scanf("%d", &a[i][j]);
    }

    printf("Enter elements of second matrix:\n");
    for(i = 0; i < r2; i++)
    {
        for(j = 0; j < c2; j++)
            scanf("%d", &b[i][j]);
    }

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            product[i][j] = 0;

            for(k = 0; k < c1; k++)
                product[i][j] += a[i][k] * b[k][j];
        }
    }

    printf("Product of the matrices:\n");
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
            printf("%d ", product[i][j]);

        printf("\n");
    }

    return 0;
}
```

```
Enter rows and columns of first matrix: 2
2
Enter rows and columns of second matrix: 2
2
Enter elements of first matrix:
11
22
11
22
Enter elements of second matrix:
33
44
22
44
Product of the matrices:
847 1452
847 1452
```

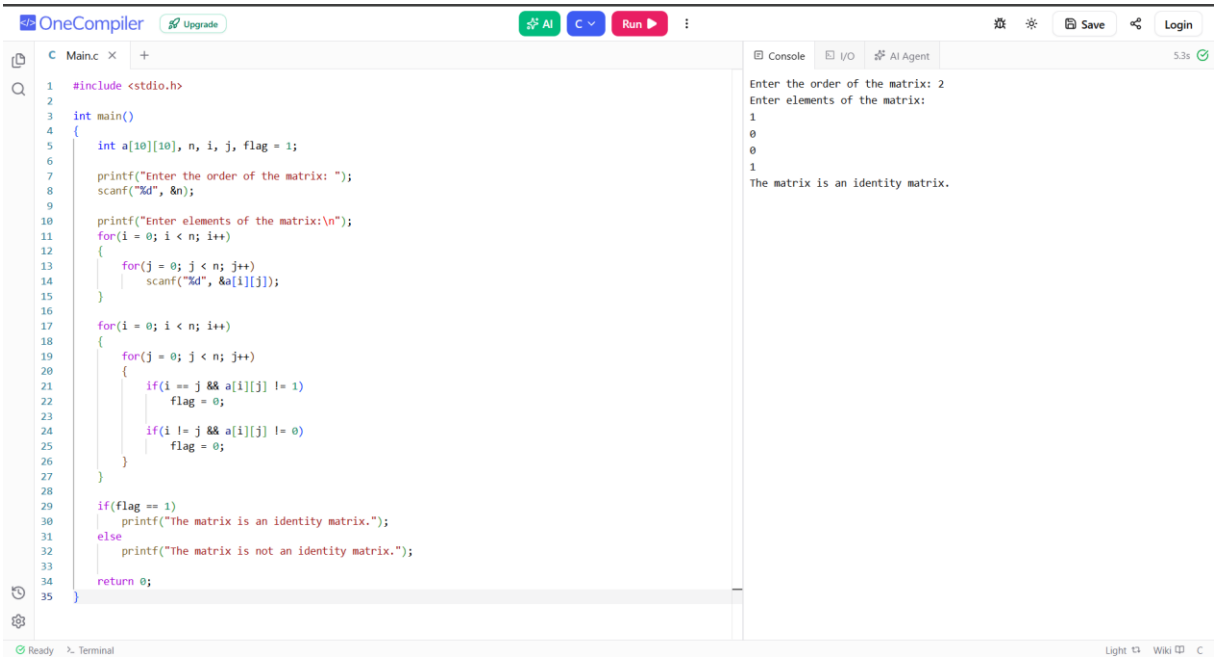
8.Matrix transpose.



9.Right diagonal sum



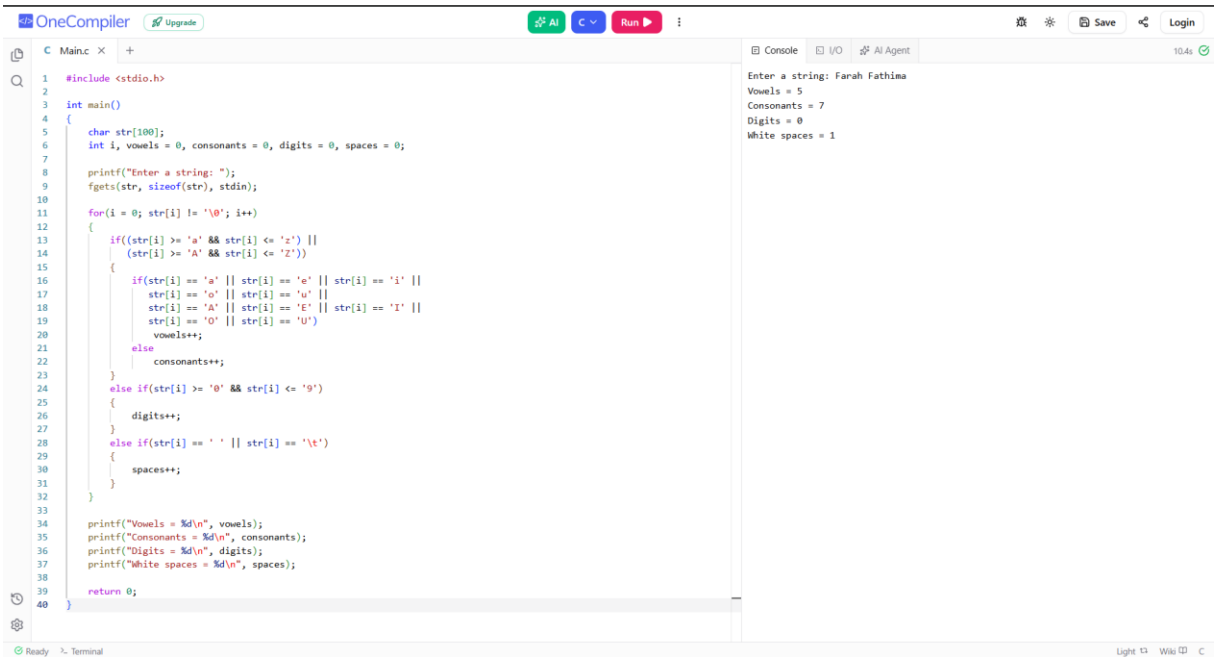
10.Check for identity matrix



```c
#include <stdio.h>

int main()
{
    int a[10][10], n, i, j, flag = 1;

    printf("Enter the order of the matrix: ");
    scanf("%d", &n);

    printf("Enter elements of the matrix:\n");
    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j && a[i][j] != 1)
                flag = 0;

            if(i != j && a[i][j] != 0)
                flag = 0;
        }
    }

    if(flag == 1)
        printf("The matrix is an identity matrix.");
    else
        printf("The matrix is not an identity matrix.");

    return 0;
}
```

```
Enter the order of the matrix: 2
Enter elements of the matrix:
1
0
0
1
The matrix is an identity matrix.
```

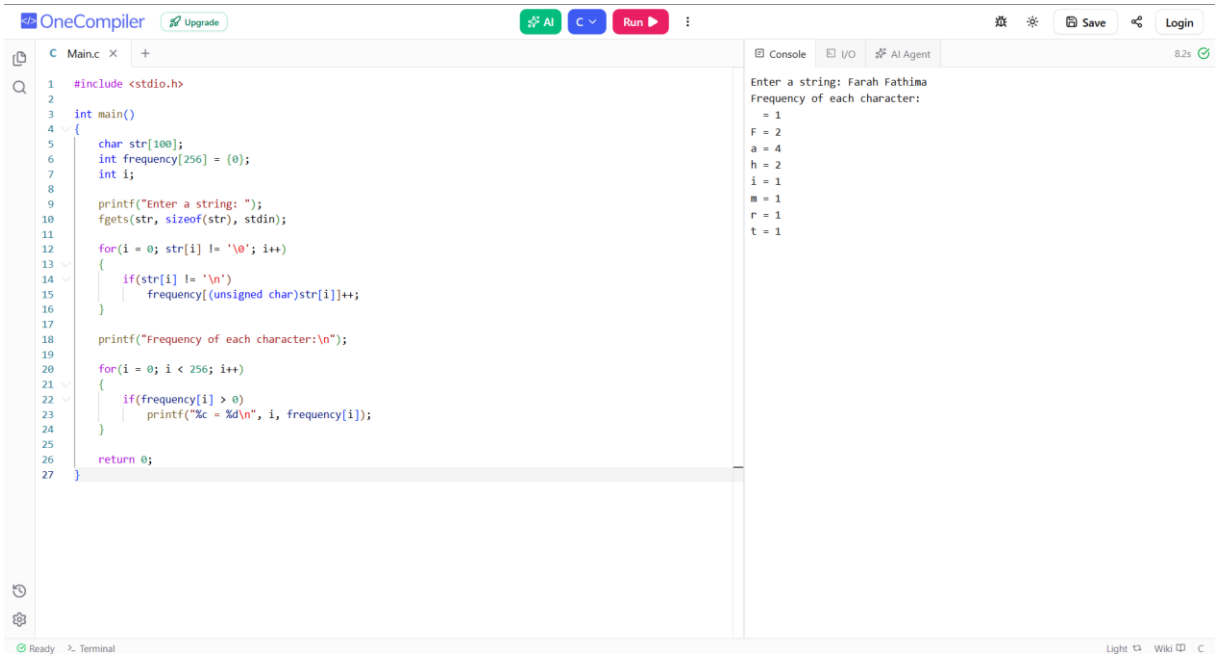
11.To find out number of vowels, consonants, digits, and white spaces



```c
#include <stdio.h>

int main()
{
    char str[100];
    int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((str[i] >= 'a' && str[i] <= 'z') ||
           (str[i] >= 'A' && str[i] <= 'Z'))
        {
            if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
               str[i] == 'o' || str[i] == 'u' ||
               str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
               str[i] == 'O' || str[i] == 'U')
                vowels++;
            else
                consonants++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(str[i] == ' ' || str[i] == '\t')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White spaces = %d\n", spaces);

    return 0;
}
```

```
Enter a string: Farah Fathima
Vowels = 5
Consonants = 7
Digits = 0
White spaces = 1
```

12. Frequency of character in a string.



```c
#include <stdio.h>

int main()
{
    char str[100];
    int frequency[256] = {0};
    int i;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] != '\n')
            frequency[(unsigned char)str[i]]++;
    }

    printf("Frequency of each character:\n");

    for(i = 0; i < 256; i++)
    {
        if(frequency[i] > 0)
            printf("%c = %d\n", i, frequency[i]);
    }

    return 0;
}
```

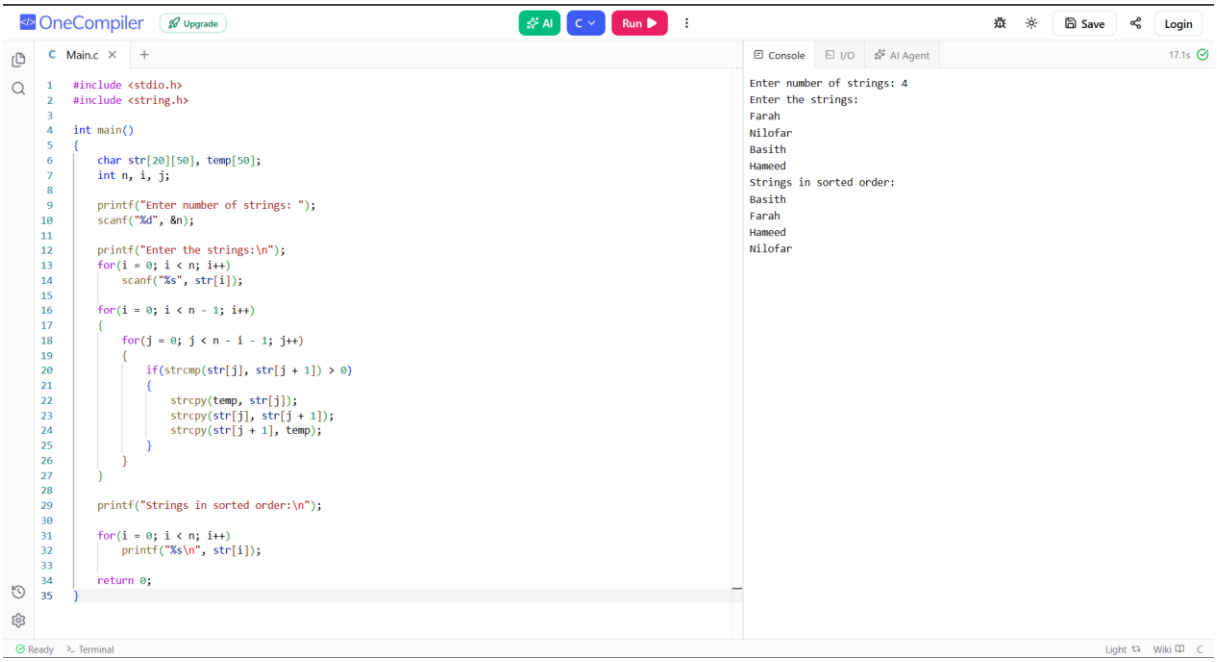
```
Enter a string: Farah Fathima
Frequency of each character:
  = 1
F = 2
a = 4
h = 2
i = 1
m = 1
r = 1
t = 1
```

13. Count words in a string

```c
#include <stdio.h>

int main()
{
    char str[200];
    int i, words = 0, inWord = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] != ' ' && str[i] != '\t' && str[i] != '\n')
        {
            if(inWord == 0)
            {
                words++;
                inWord = 1;
            }
        }
        else
        {
            inWord = 0;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}
```

```
Enter a string: Farah Fathima
Number of words = 2
```

14.Sort string array



15.Toggle case of a sentence.